

On the Impossibility of Informationally Efficient Markets

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ABSTRACT

Grossman and Stiglitz ask whether a competitive market can ever be informationally efficient in the strong sense that prices fully reveal all the information investors would otherwise pay to acquire. Building a noisy rational expectations model with constant absolute risk aversion and normally distributed payoffs, they show that whenever information is costly, full revelation is self-defeating: if price already told everyone what informed traders know, no one would pay to become informed, the price would then carry no information, and the assumed equilibrium would unravel. The paper's constant-absolute-risk-aversion example yields a closed-form equilibrium in which an endogenous fraction of traders acquires a costly private signal, and the resulting price is a linear, noisy aggregator of that signal, with random per-capita supply providing the noise that keeps the system from becoming fully revealing. The authors derive closed-form comparative statics for the equilibrium fraction informed and for the informativeness of price, and prove that a competitive equilibrium fails to exist in the two limiting cases of zero noise and costlessly perfect information. The result reframes, rather than refutes, the efficient markets hypothesis: efficiency and the private incentive to produce information are jointly determined and mutually constraining, not independent properties of a market.

1. Introduction

By 1980 the efficient markets hypothesis, formalized a decade earlier by Fama (1970) — already reviewed in this library under the Financial Markets series — had become the organizing proposition of academic finance. Its central claim, that security prices "fully reflect" all available information, had passed a long sequence of empirical tests (serial correlation, filter rules, event studies, mutual fund performance) with results that were, on the whole, remarkably favorable to the hypothesis. Grossman and Stiglitz do not run a new test. They ask a prior, logical question that the empirical literature had largely stepped around: if prices really did fully reflect all available information, including information that is costly to acquire, what would induce anyone to bear that cost in the first place? And if no one bears the cost, on what basis would prices reflect the information at all?

The paper answers this question with a fully worked model rather than an informal argument. It constructs the simplest possible economy in which price can, in principle, transmit information from informed to uninformed traders — a single risky asset, constant absolute risk aversion, normally distributed payoffs and supply — and shows that the logical tension identified above is not a curiosity of extreme cases but a structural property of the equilibrium: a market that is fully informationally efficient in Fama's strong sense cannot simultaneously be a competitive equilibrium in which information is costly. The two limiting cases that would make the market fully efficient (zero noise in supply, or perfectly precise information) are exactly the two cases in which no competitive equilibrium exists at all. The paper's title is therefore to be read literally, and narrowly: not "markets are never efficient," but "the polar case of perfect informational efficiency is impossible to sustain as an equilibrium once information is costly."

2. Research Problem and Hypothesis

2.1 Research Question

The paper asks a precisely defined question rather than a general one about "market efficiency": in an economy in which a costly private signal about a risky asset's payoff can be acquired by any trader, and in which price is set competitively so as to clear the market for the asset, does there exist an equilibrium fraction of informed traders, and an associated equilibrium price function, such that (i) each trader is indifferent, at the margin, between paying the cost of information and remaining uninformed, and (ii) the resulting price is exactly as informative as that joint decision implies — and, crucially, can this equilibrium ever coincide with the case in which price reveals the informed traders' signal perfectly?

2.2 Motivation

The question matters because the efficient markets hypothesis, taken to its logical extreme, is internally inconsistent with the existence of the very activity — costly information production and arbitrage — that is usually invoked to explain why prices are informative at all. If informed trading is what makes prices informative, and informed trading requires a return sufficient to cover the cost of information, then a price that already impounded all information would leave that return at zero, extinguishing the incentive that produced the informativeness in the first place. This is not a minor technical caveat; it goes to the foundations of how economists justify the informational role of the price system, a role traced by the authors to Hayek (1945) and formalized in the rational expectations tradition of Lucas (1972).

2.3 Existing Literature

The paper is explicit about its lineage. It extends the noisy rational expectations framework introduced by Lucas (1972) and applied to information flows between traders by Green (1973) and by Grossman (1975, 1976, 1978), and by Kihlstrom and Mirman (1975). That earlier literature had already established that price can act as an imperfect aggregator of dispersed private information when some exogenous source of noise (in Grossman's work, unpredictable asset supply arising from hedging or storage decisions) prevents price from being a fully revealing statistic. What that literature had not done, prior to this paper, is close the model on the demand side for information itself — making the fraction of informed traders an equilibrium object determined jointly with the informativeness of price, rather than a parameter fixed by the modeler.

2.4 Research Gap

The specific gap is the absence, at the time of writing, of a model in which (a) information is explicitly costly, (b) the decision to acquire it is made by utility-maximizing, price-taking agents, and (c) the resulting equilibrium is characterized in closed form so that the tension between informational efficiency and the incentive to produce information can be examined analytically rather than asserted informally. Green (1977) had shown, in a related setting, that fully revealing rational expectations equilibria need not exist; Grossman and Stiglitz cite this result but push further, asking not merely whether a fully revealing equilibrium exists, but what equilibrium looks like when information is costly and traders correctly anticipate how informative price will be before they decide whether to pay for information.

2.5 Contribution

PAPER The authors claim three contributions: a general set of seven conjectures about the qualitative properties of any such equilibrium (Section I of the paper); a fully solved constant-absolute-risk-aversion, normal-variables example in which every conjecture is verified in closed form (Sections II–III); and a reinterpretation of the efficient markets hypothesis showing that costless information is a sufficient, but not a necessary, condition for prices to fully reflect available information — and that the necessary condition (positive information cost) is precisely what prevents full revelation from being sustainable (Section IV).

CRITIQUE The claimed contribution is substantially demonstrated for the constant-absolute-risk-aversion, normal-variable example, where every one of the seven conjectures is proven as a formal theorem. It is not demonstrated for the general model sketched in Section I; the authors are candid about this, stating outright that they "have not been able to obtain a general proof of any of these propositions" outside the tractable example. The paper's generality claims should therefore be read as conjectures illustrated by a fully worked special case, not as general theorems.

3. Theoretical Framework and Mechanism

3.1 Assets, Endowments, and the Information Structure

There are two assets: a safe asset paying a fixed return R per dollar invested, and a risky asset whose payoff at the end of the period, $\tilde{u}$, is random. The payoff decomposes as

$$\tilde{u} = \theta + \varepsilon \quad (1)$$

θ the component of the risky payoff that can be learned, at a fixed cost c , by any trader who chooses to pay for it.

ε a residual, unobservable to everyone, uncorrelated with θ , representing the part of the payoff no amount of costly research can resolve.

θ and ε are jointly normal, independent, with $E\varepsilon = 0$, $Cov(\theta, \varepsilon) = 0$, and $Var(\tilde{u}|\theta) = Var \varepsilon = \sigma_\varepsilon^2 > 0$ (equations 4–6 of the paper). A fraction λ of traders chooses, in equilibrium, to pay c and observe θ before trading (the informed); the remaining $1-\lambda$ observe nothing but the price itself (the uninformed). Every trader is ex ante identical — informed status is purely a choice, not an endowment — which is what makes λ an equilibrium object rather than a primitive of the model. Per-capita supply of the risky asset, $\bar{x}$, is itself random and unobserved by the uninformed; it is this supply noise, not any assumed cognitive limitation, that prevents the uninformed from inferring θ exactly from price.

3.2 Individual Optimization and Asset Demand

Each trader has exponential (constant absolute risk aversion, "CARA") utility over end-of-period wealth, $V(W) = -e^{-aW}$, $a > 0$. Given a portfolio of M dollars in the safe asset and X units of the risky asset, end-of-period wealth is

$$W_1^* = RM + \tilde{u}X \quad (3)$$

subject to the budget constraint $PX + M = W_0$, where W_0 is initial wealth marked to the current price P of the risky asset (equation 2). Because W_1^* is normal conditional on whatever the trader knows, maximizing $E[V(W_1^*)]$ reduces to a mean-variance problem, and the CARA-normal

combination delivers demand functions that do not depend on wealth (so the model can be solved without tracking the cross-sectional wealth distribution). For an informed trader who has observed θ :

$$X_I(P, \theta) = (\theta - RP) / (a\sigma_\varepsilon^2) \quad (8)$$

The informed trader treats θ as known, so the only risk remaining in $\tilde{u}$ is ε ; demand is the familiar mean-variance ratio of the perceived risk premium $(\theta - RP)$ to risk aversion times residual variance. For an uninformed trader, who observes only price and must use it, together with rational expectations about the equilibrium price function $P_\lambda(\cdot)$, to form beliefs about $\tilde{u}$:

$$X_U(P; P_\lambda) = (E[\tilde{u} | P_\lambda = P] - RP) / (a \cdot \text{Var}[\tilde{u} | P_\lambda = P]) \quad (8')$$

Both demands have the same mean-variance form; the entire difference between an informed and an uninformed trader is that the uninformed must condition on price rather than on θ directly, and price is a noisier signal of θ than θ is of itself. This is the single mechanism through which the model operates: the two trader types hold formally identical objective functions and differ only in the information set that enters the conditional moments of $\tilde{u}$.

3.3 The Rational-Expectations Price Function

Given a fraction λ of informed traders, an equilibrium price function $P_\lambda(\theta, x)$ must clear the market for every realization of (θ, x) :

$$\lambda \cdot X_I(P_\lambda(\theta, x), \theta) + (1 - \lambda) \cdot X_U(P_\lambda(\theta, x); P_\lambda) = x \quad (9)$$

The uninformed cannot separately observe θ and x , only the single number $P_\lambda(\theta, x)$; because a high price could reflect either high θ or low supply, price transmits information about θ only up to the noise contributed by x . Grossman and Stiglitz show that the equilibrium price is informationally equivalent to a single composite signal:

$$w_\lambda(\theta, x) = \theta - (a\sigma_\varepsilon^2/\lambda)(x - E\tilde{x}), \quad \lambda > 0 \quad (10a)$$

a "mean-preserving spread" of θ whose noise term is proportional to the deviation of supply from its mean and inversely proportional to λ : the fewer traders are informed, the more the informed traders' aggregate demand has to move to absorb a given supply shock, and the noisier is the signal it embeds in price. **PAPER Theorem 1** establishes that a linear equilibrium of this form exists and is unique in the class of price functions monotone in w_λ : $P_\lambda(\theta, x) = a_1 + a_2 w_\lambda(\theta, x)$ with $a_2 > 0$, the closed form of the coefficients given in the paper's Appendix B. The residual uncertainty the uninformed face after observing price is exactly the conditional variance of the noise term:

$$\text{Var}[w_\lambda | \theta] = (a\sigma_\varepsilon^2/\lambda)^2 \cdot \text{Var}(\tilde{x}) \quad (11)$$

3.4 Equilibrium in the Market for Information

An overall equilibrium is a pair (λ, P_λ) such that no trader wants to switch information status: if $0 < \lambda < 1$, the expected utility of being informed must equal that of being uninformed, net of the cost c ; corner solutions ($\lambda=0$ or $\lambda=1$) apply when one side always dominates. **PAPER Theorem 2** derives the ratio of the two expected utilities in closed form:

$$EV(W_I) / EV(W_U) = e^{ac} \cdot \sqrt{Var(\tilde{u}|\theta) / Var(\tilde{u}|w_\lambda)} \equiv \gamma(\lambda) \quad (13-14)$$

The exponential term e^{ac} captures the direct utility cost of paying c ; the square-root term captures the informational advantage of knowing θ exactly relative to learning only the noisy composite w_λ . **PAPER Corollary 1** shows $\gamma(\cdot)$ is strictly monotone increasing in λ (because utility is expressed as a negative number, a rising ratio means informed advantage is *falling* as more traders become informed — more informed traders make price more revealing, which erodes precisely the private edge that justified paying c). **PAPER Theorem 3** then establishes existence and uniqueness of an interior equilibrium $\lambda^* \in (0,1)$ solving $\gamma(\lambda)=1$ whenever $\gamma(0) < 1 < \gamma(1)$, with corner solutions otherwise. This is the equilibrium determination illustrated in the paper's own Figure 1, reconstructed in Section 5 below.

3.5 Characterizing the Informativeness of Price

To make the comparative statics tractable, the authors define two composite parameters:

$$m \equiv (a\sigma_\varepsilon^2/\lambda)^2 \cdot Var(\tilde{x})/\sigma_\theta^2 \quad n \equiv \sigma_\theta^2/\sigma_\varepsilon^2 \quad (16a-16b)$$

m is inversely related to how informative price is — it is the ratio of the noise variance injected into the price signal to the prior variance of θ — while n measures how much of the risky payoff's total variance is, in principle, learnable (the signal-to-noise ratio of θ itself within $\tilde{u} = \theta + \varepsilon$). The squared correlation between price and θ , the natural summary measure of price informativeness, is

$$\rho_{p\theta}^2 = 1/(1+m) \quad (17)$$

and, using the standard normal-updating formula for conditional variance under a Gaussian signal, the ratio in equation (14) reduces to a function of m and n alone:

$$Var(\tilde{u}|\theta) / Var(\tilde{u}|w_\lambda) = (1+m) / (1+m+nm) \quad (18)$$

Substituting into the equilibrium condition $\gamma(\lambda)=1$ collapses the entire equilibrium characterization to a single scalar equation:

$$e^{2ac} = 1 + nm/(1+m) \quad (19a-19b)$$

This is, in the reviewer's judgment, the paper's single most important analytical result, because it says something that is not obvious from the setup: *in equilibrium, the informativeness of price (summarized by m , and hence ρ^2) is pinned down entirely by the cost of information c , the quality of the informed signal n , and risk aversion a — and not by the level of noise $\text{Var}(\tilde{x})$ or the total variance of θ taken separately.* Any change in the amount of noise, or in the overall scale of uncertainty, is exactly offset by an endogenous adjustment in λ , leaving equilibrium informativeness unchanged. This is **PAPER Theorem 4**, whose four comparative-statics results are: (A) equilibrium informativeness rises if n rises, c falls, or a falls; (B) equilibrium informativeness is *unchanged* by a change in $\text{Var}(\tilde{x})$ alone, or by a change in $\text{Var}(\tilde{x})$ holding n fixed; (C) the equilibrium fraction informed rises if $\text{Var}(\tilde{x})$ rises (with n free or fixed) or c falls; (D) the effect of n on λ itself is ambiguous and depends on whether n is above or below a threshold $\bar{n}$ defined implicitly by the paper's footnote 8.

3.6 The Impossibility Result

The paper's title result follows directly from equation (19) by examining its two degenerate limits. **PAPER Theorem 5:** (a) if there is no supply noise ($\text{Var}(\tilde{x})=0$), an overall equilibrium fails to exist whenever $c^a < \sqrt{1+n}$; (b) if information is perfectly precise ($\sigma_\varepsilon^2=0$, i.e., $n \rightarrow \infty$), an equilibrium *never* exists, for any $c > 0$. The proof is a discontinuity argument: at these limits the ratio $\gamma(\lambda)$ jumps as λ crosses zero — at $\lambda=0$ price is completely uninformative (so becoming informed is strictly attractive, $\gamma(0) < 1$), yet for any $\lambda > 0$ price becomes instantly and perfectly revealing (so being informed is strictly dominated once anyone else is, $\gamma(\lambda) > 1$). No fraction λ can simultaneously satisfy both boundary conditions, so no fixed point exists: this is the impossibility of a fully informationally efficient competitive equilibrium once information costs anything at all. Economically, the mechanism is exactly the intuitive one that motivates the paper's title, now made rigorous: full revelation and a positive private return to information cannot coexist, and the model shows there is no escape route through parameter values — the contradiction is structural, not a matter of degree.

4. Results: What the Theorems Establish

THEOREM 1 — EXISTENCE AND LINEARITY OF THE PRICE FUNCTION

For any $\lambda \in (0, 1]$, a competitive equilibrium price exists and is a linear function of the composite signal w_λ . Economically: the price mechanism does exactly what a "well articulated" information-transmission device should do — it is neither degenerate nor pathological for any interior fraction of informed traders. This is a necessary preliminary for everything that follows, but it is not, by itself, a statement about efficiency.

THEOREM 3 — EXISTENCE AND UNIQUENESS OF THE EQUILIBRIUM FRACTION INFORMED

Given the monotonicity of $\gamma(\lambda)$ (Corollary 1), there is exactly one equilibrium λ^* , interior whenever the utility ratio brackets unity between the two corners. Economically: the market for information itself, not merely the market for the asset, clears at a unique point, so the model delivers a determinate prediction rather than a set of possible outcomes.

THEOREM 4 — COMPARATIVE STATICS OF INFORMATIVENESS

Equilibrium price informativeness depends only on (c, n, a) ; changes in the level of noise or in overall payoff variance are absorbed by an offsetting change in λ^* and leave informativeness unchanged. This is the paper's most economically surprising finding: two markets with very different amounts of "noise trading" can exhibit identical price informativeness in equilibrium, because the market recruits exactly as many informed traders as are needed to restore the same informational content. The illustrative calculation in Section 5 makes the magnitude of this offsetting adjustment concrete.

THEOREM 5 — NONEXISTENCE AT THE EFFICIENT-MARKET LIMIT

No competitive equilibrium exists when noise vanishes or information becomes perfect. This is the paper's central and most cited result. It is a genuine impossibility theorem, not a claim about how close markets can get to full efficiency; it identifies the boundary case that a competitive equilibrium can approach but can never reach, given any positive cost of information.

THEOREM 6 — VANISHING TRADE AT THE SAME LIMITS

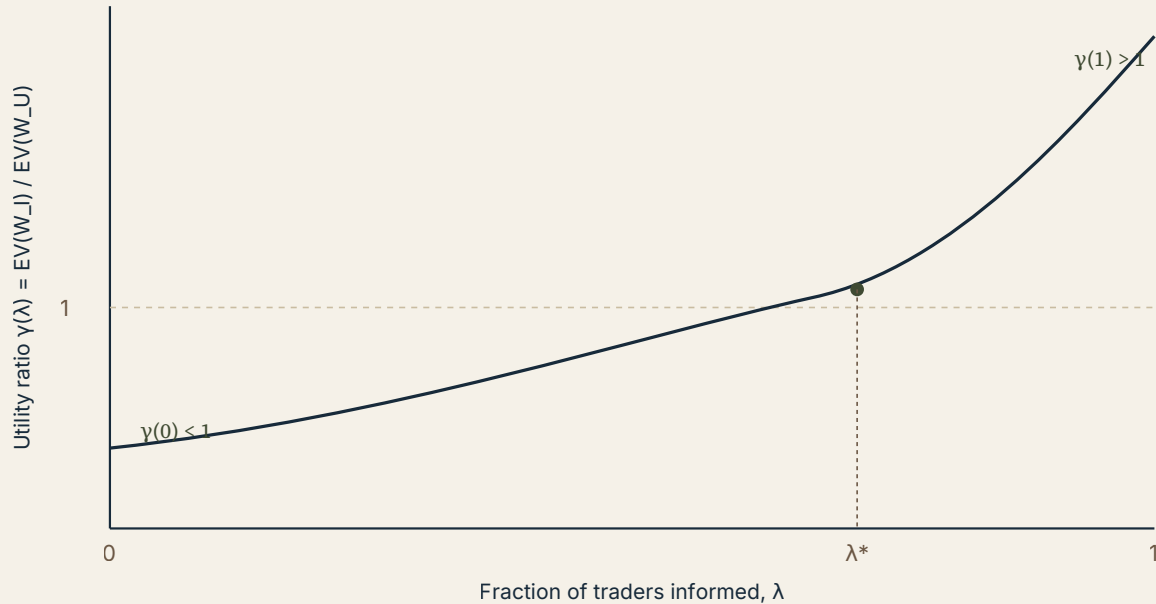
As the model approaches either degenerate limit, both the mean and the variance of trading volume go to zero (Section III, discussed in Section 6 below). This is offered by the authors as the economic resolution of the paradox: markets do not actually reach the impossible limit because they cease to function — through vanishing volume — before they get there.

None of these results carries an explicit measure of statistical or economic "significance" in the empirical sense, since the paper contains no estimation. Their economic significance instead lies in the qualitative shape of the equilibrium they describe: prices are informative, but only partially and only because they are not too informative; the very noise that a naive reading of market efficiency would treat as a nuisance is what allows a competitive market for information to have a determinate, non-degenerate solution at all.

5. Table and Figure Analysis

5.1 Figure 1 — Determination of the Equilibrium Fraction Informed

Figure 1. Schematic reconstruction of the equilibrium condition $\gamma(\lambda) = 1$



Legend: schematic reconstruction by the reviewer of the qualitative shape described analytically by the paper's Theorem 3 and Corollary 1 (monotone increasing $\gamma(\lambda)$ crossing unity at a unique interior λ^*); the original AER printing of Figure 1 was not available for direct reproduction. Axes are illustrative, not to scale. Source: authors' construction, based on Grossman and Stiglitz (1980), Section II.F–G.

The figure answers a single question: given the parameters (a, c, n) , where does the private incentive to acquire information exactly balance the cost of acquiring it? Because $\gamma(\cdot)$ is proved strictly monotone, the crossing point is unique, and everything to the left of λ^* represents a fraction of informed traders too small for equilibrium (informed traders still earn a premium, drawing in more entrants) while everything to the right represents a fraction too large (the informed premium has been competed away and traders would rather free-ride on price). The diagram is a description of an equilibrium condition, not of any observable relationship in real data; there is no causal claim to evaluate here beyond the internal consistency of the fixed point.

5.2 Illustrative Comparative-Statics Calculation

EXTENSION The following table is not in the paper. It is a numerical illustration, computed by the reviewer, of Theorem 4's comparative statics, obtained by solving equations (16a) and (19a) in closed form for λ^* and ρ^2 given assumed parameter values. Because equation (19a) can be inverted analytically — $m = (e^{2ac} - 1) / (n - (e^{2ac} - 1))$, then $\lambda^* = a\sigma_\varepsilon^2 \sqrt{Var(\tilde{x}) / (\sigma_\theta^2 m)}$ from (16a) — no simulation is required; the values below are exact given the assumptions stated. Assumptions used for this illustration only: risk aversion $a=1$, residual variance $\sigma_\varepsilon^2=1$, supply-noise variance $Var(\tilde{x})=0.01$ (a "thin noise trading" calibration chosen only so that λ^* falls inside the economically meaningful unit interval; these are illustrative magnitudes, not estimates of any real market).

Table 1. Equilibrium fraction informed and price informativeness under Theorem 4 (illustrative calculation)

Signal quality, $n = \sigma_\theta^2 / \sigma_\varepsilon^2$	Information cost, c	Equilibrium λ^*	Price informativeness, ρ^2
1 (low quality)	0.010	0.696	0.980
1 (low quality)	0.030	0.390	0.938
1 (low quality)	0.060	0.262	0.873
4 (moderate quality)	0.010	0.702	0.995
4 (moderate quality)	0.030	0.399	0.985
4 (moderate quality)	0.060	0.276	0.968
16 (high quality)	0.010	0.703	0.999
16 (high quality)	0.030	0.401	0.996
16 (high quality)	0.060	0.279	0.992

Source: reviewer's calculation, solving equations (16a) and (19a) of Grossman and Stiglitz (1980) in closed form. Not an official result of the paper; presented for illustration of Theorem 4 only.

Reading down each block (fixed n , rising c): the equilibrium fraction informed collapses quickly — roughly two-thirds to one-quarter over this range of costs — exactly as Theorem 4(C) predicts, and informativeness falls with it, exactly as Theorem 4(A) predicts. Reading across each row (fixed c , rising n): the equilibrium λ^* is nearly invariant to signal quality (it moves by less than one percentage point as n rises sixteen-fold), while informativeness rises steadily toward unity. This is the compensating-adjustment result of Theorem 4(B) rendered concrete: because these parameter values sit in the region where the threshold condition of footnote 8 makes $\partial\lambda/\partial n$ small and slightly negative, almost all of the benefit of better-quality private information shows up as higher price informativeness rather than as a larger informed population — the market absorbs improved signal quality mainly through more revealing prices, not through recruiting more researchers.

6. Robustness, Limiting Cases, and the Authors' Own Extensions

The paper contains no empirical robustness section in the conventional sense; its robustness checks are analytical — limiting cases and alternative model features explored to see whether the impossibility result survives. Each is evaluated below using the same concern-test-result logic CLAUDE.md prescribes for empirical robustness sections, adapted to a theoretical setting.

CONCERN: IS THE IMPOSSIBILITY RESULT AN ARTIFACT OF THE EXACT ZERO-NOISE LIMIT, NEVER ACTUALLY APPROACHED?

Test (Section III of the paper): the authors compute the mean and variance of equilibrium trading volume as a function of the exogenous parameters and examine the same three limits used in Theorem 5 (cost going to the threshold at which $\lambda=1$, signal precision $n \rightarrow \infty$, and noise $\text{Var}(\tilde{x}) \rightarrow 0$).

Result (Theorem 6): in all three limits, both the mean and the variance of per-capita net trade go to

zero. **Does it resolve the concern?** EVIDENCE Only partially. It shows that the model is internally consistent — markets do not approach the impossible limit with high volume and functioning liquidity, which would have been a much sharper paradox; instead, they thin out and effectively cease to operate as the limit is approached. CRITIQUE But "trade goes to zero" is not the same as "an economically sensible story for why markets close." The result is a mathematical consequence of traders' beliefs converging (near the limits, informed and uninformed traders come to hold nearly identical beliefs, eliminating the divergence in beliefs that motivates trade in this model), not an account of any real-world friction, entry cost, or market-maker decision that would actually cause a market to shut down.

CONCERN: DOES THE IMPOSSIBILITY DEPEND ON TRADING BEING ANONYMOUS AND NON-STRATEGIC (PRICE-TAKING)?

Test: the paper does not test this directly; it assumes price-taking (competitive, not strategic) behavior throughout, citing Green (1977) and Stiglitz (1971, 1974) on related non-existence results under similar competitive assumptions. **Result:** no alternative (strategic) equilibrium concept is solved in this paper. **Does it resolve the concern?** CRITIQUE No — this is a genuine open flank of the paper, addressed five years later by Kyle (1985), whose strategic, imperfectly competitive informed monopolist generates a fully determinate, always-existing equilibrium with a continuous trade-off between informativeness and profitability, without any impossibility result. The comparison across the two papers (Section 11 below) suggests the nonexistence result is at least partly an artifact of the price-taking assumption rather than an unavoidable feature of costly information per se.

CONCERN: DOES THE ARGUMENT DEPEND ON THE SPECIFIC FUNCTIONAL FORMS (CARA UTILITY, NORMAL PAYOFFS, SINGLE RISKY ASSET)?

Test: the paper offers Section I's seven conjectures as claims meant to hold beyond the CARA-normal example, but explicitly does not prove them in general. A footnote also notes that heterogeneous risk aversion across traders does not change any Section II result. **Result:** generality is asserted for the conjectures, proved only for the tractable case; robustness to heterogeneous risk aversion is asserted and briefly justified but not derived step by step in the text. **Does it resolve the concern?** CRITIQUE Not fully. The functional-form dependence is a standard and defensible modeling choice for obtaining closed-form results, but it does mean Theorem 5's exact algebraic threshold ($\epsilon^{ac} < \sqrt{(1+n)}$) is specific to CARA-normal; what is genuinely general, and what later papers (Hellwig 1980; Diamond and Verrecchia 1981; Admati 1985) confirm in richer settings, is the qualitative impossibility of a costly-information, fully-revealing competitive equilibrium, not the precise functional threshold.

CONCERN: WHAT HAPPENS IF ADDITIONAL SECURITIES ARE INTRODUCED TO COMPLETE THE MARKET?

Test (Section IV): the authors informally analyze adding a binary security paying off on the event $\tilde{u} > E\theta$, arguing that under weak conditions its price, together with the original asset's price, would jointly reveal θ almost completely. **Result:** the authors argue such a market could not be an equilibrium for the same reason as Theorem 5 — a "noiseless" market again removes any return to information. **Does it resolve the concern?** CRITIQUE This extension is argued informally, not proven

with the same rigor as Theorems 1–6; it is best read as a suggestive conjecture ("there cannot be as many securities as states of nature" under costly information) rather than an established result of the paper.

7. Critical Evaluation: What Does the Paper Actually Prove?

ESTABLISHED

Within the CARA-normal example: existence, uniqueness, and linearity of the informative price function for any interior λ (Theorem 1); existence and uniqueness of the equilibrium fraction informed (Theorem 3); the closed-form comparative statics of Theorem 4, including the compensating-adjustment result that equilibrium informativeness is invariant to noise levels holding (a, c, n) fixed; and the nonexistence of a competitive equilibrium at the two degenerate limits of zero noise and perfect information (Theorem 5). These are formally proved, with proofs supplied in the paper's appendices, and are not contingent on any data or estimation.

SUPPORTED BUT CONDITIONAL

The claim that markets "thin out" as they approach the impossible limit (Theorem 6) is established within the model but is conditional on the specific belief-divergence channel that generates trade in this economy; it is not shown to be the mechanism behind observed illiquidity in any actual thin market. The claim that heterogeneous risk aversion does not affect the results is asserted with a one-line justification rather than derived, and should be treated as conditional on that (plausible but unverified in the text) claim.

SUGGESTIVE

The seven conjectures of Section I, beyond those subsequently proved as theorems in the CARA-normal case, are suggestive generalizations rather than established results: the authors state directly that no general proof was obtained. The Section IV discussion of adding securities to "complete" the market and the resulting further impossibility is likewise suggestive and informal, not a proved theorem of the paper.

NOT ESTABLISHED

The paper establishes nothing about the *welfare* properties of the equilibrium it characterizes; the authors state explicitly, in the final paragraph, that they "have said nothing regarding the social benefits of information, nor whether it is socially optimal to have 'informationally efficient markets,'" deferring this to future work. It also establishes nothing empirically — there is no test of whether real markets exhibit the specific comparative statics of Theorem 4, and no calibration of (a, c, n) to any actual security. Finally, the paper does not establish that fully strategic, price-impact-aware informed trading would generate the same impossibility; that question is left open and answered differently by later work (Section 11).

8. Strengths

STRENGTHS RECOGNIZED BY THE AUTHORS

The authors emphasize, correctly, that their model is "the simplest one in which prices perform a well-articulated role in conveying information," and that the closed-form CARA-normal example allows every qualitative conjecture from the general discussion to be checked exactly rather than asserted. They also flag, appropriately, that they are attempting to "redefine the Efficient Markets notion, not destroy it" — correctly anticipating that the paper would otherwise be read as a wholesale refutation of Fama (1970) rather than as a refinement of its scope conditions.

STRENGTHS IDENTIFIED INDEPENDENTLY

The paper's most durable strength is analytical economy: a two-parameter reduction (m and n) collapses a multi-dimensional fixed-point problem into a single scalar equation (19a), from which every comparative static in Theorem 4 follows by inspection. This tractability is precisely what allowed the paper to spawn an entire subsequent literature (Hellwig 1980; Diamond and Verrecchia 1981; Verrecchia 1982; Admati 1985) that varies the number of assets, signals, and traders while preserving the same CARA-normal machinery. A second, less remarked strength is that the paper's central point is architectural rather than parametric: it is not that markets are "usually" inefficient by some degree that could in principle shrink to zero with better technology; it is that the fully efficient case is a measure-zero limit that removes its own precondition. That distinction — a logical impossibility rather than an empirical tendency — is what has kept the paper's citation relevant across four subsequent decades of changing information technology, since no reduction in the physical cost of information can, by the model's own logic, eliminate a positive c and hence eliminate the mechanism.

9. Weaknesses and Critique

Problem: price-taking, non-strategic informed traders. **Why it matters:** a single informed trader in this model behaves as if his own trade has no effect on price, yet the aggregate of informed trades is precisely what makes price informative; this is the standard "competitive" fiction pushed to a setting where it does real analytical work. **Potential bias:** it may overstate how sharply the impossibility bites, since a strategic informed trader (as in Kyle 1985) internalizes the effect of his own trade on how much he reveals, and can therefore profitably trade even when, in aggregate, prices are highly informative. **How it could be tested:** re-solve the model with a finite number of Cournot-Kyle strategic informed traders and check whether the discontinuity underlying Theorem 5 survives as the number of strategic traders grows large.

Problem: the source of trade is restricted to belief differences generated purely by the information asymmetry itself. **Why it matters:** the authors deliberately shut down preference and endowment heterogeneity ("this paper focuses on the last of these three [sources of trade]"), so all of the model's predictions about volume, thinness, and market breakdown are conditional on that restriction. **Potential bias:** real markets have persistent endowment and preference heterogeneity

that would sustain trade and possibly liquidity even in the model's degenerate limits, softening or eliminating the "vanishing volume" resolution of Theorem 6. **How it could be tested:** re-introduce heterogeneous endowments (which the authors flag but exclude in footnote 9 as adding "undue complication") and check whether trade volume remains bounded away from zero as $Var(\tilde{x}) \rightarrow 0$ or $n \rightarrow \infty$.

Problem: a continuum of ex ante identical traders choosing a continuous fraction λ is a mathematical convenience with no discrete microfoundation. **Why it matters:** in any finite market, λ must be a ratio of integers, and the entry/exit decision of the marginal trader is lumpy, not continuous; the model's smooth equilibrium condition $y(\lambda)=1$ presumes away integer constraints that could matter in small or concentrated markets. **Potential bias:** in markets with few potential informed traders (a niche security, a small emerging market), the continuous approximation could mask genuinely discrete, multiple-equilibrium behavior. **How it could be tested:** a discrete-agent version of the model, solved for small trader counts, checked against the continuum limit as the number of traders grows.

CRITIQUE The single most important weakness, in the reviewer's judgment, is the first one: the assumption of price-taking behavior by informed traders is doing more analytical work than the paper's framing acknowledges. The paper presents the nonexistence result as an unconditional structural feature of "competitive markets," but Kyle's (1985) subsequent strategic model shows that a monopolistic informed trader who correctly perceives his own price impact generates a perfectly well-defined equilibrium at every level of noise, including the zero-noise limit that Grossman and Stiglitz show to be impossible under price-taking. The impossibility is therefore better described as a property of the perfectly-competitive-informed-trader assumption interacting with costly information, not as an unconditional property of "markets" as the paper's rhetoric sometimes implies.

10. Contradictions and Edge Cases

INFERENCE Under what conditions should the mechanism disappear? If information cost $c \rightarrow 0$ while remaining strictly positive, the model shows $\lambda \rightarrow 1$ at a finite threshold cost $\bar{c}$ (the remark following Theorem 4 in Section II.I), not asymptotically; below $\bar{c}$, everyone is informed and the model's information market shuts down as a corner solution, not because the impossibility mechanism was overcome but because the interior equilibrium ceased to bind. This is a real edge case the paper handles cleanly, but it is worth noting explicitly: the model never actually reaches full information at $c=0$ exactly, since the paper does not analyze that boundary case directly (it is covered only by Theorem 5(b), the perfect-information limit, which is about $n \rightarrow \infty$, not $c \rightarrow 0$).

INFERENCE Extreme values of risk aversion. As $a \rightarrow 0$ (risk neutrality), informed demand $X_I = (\theta - RP) / (a\sigma_\epsilon^2)$ becomes unboundedly sensitive to any perceived mispricing, which by Theorem 4(A) drives equilibrium informativeness toward its maximum for given (c, n) ; taken to the limit, this reproduces a version of the same impossibility, since infinitely elastic informed demand makes

price fully revealing regardless of noise. This is consistent with, rather than a violation of, Theorem 5, but the paper does not discuss the risk-neutral limit explicitly, leaving it to the reader to confirm the boundary case is not a hidden counterexample.

INFERENCE **Could another mechanism generate the same qualitative pattern?** Yes: models of costly search or rational inattention (a literature descending from Sims, not cited by Grossman and Stiglitz since it postdates the paper) generate a similarly self-limiting relationship between the cost of attention and the informativeness of prices, without requiring the specific noisy rational expectations apparatus used here. This does not contradict Grossman-Stiglitz, but it does mean the "impossibility of full efficiency under costly information" is a more general proposition than the specific CARA-normal derivation might suggest — a point in the paper's favor, but one the paper itself does not make.

CRITIQUE **Does the conclusion hold outside the model's competitive-market setting?** Not straightforwardly. In markets with a monopolistic market maker or specialist (rather than the competitive, anonymous price-taking assumed here), the market maker's bid-ask spread — rather than the fraction of informed traders — can be the margin that adjusts to compensate informed traders, a mechanism explored later by Glosten and Milgrom (1985) and Kyle (1985), neither of which produces a nonexistence result of the Grossman-Stiglitz type. The paper's impossibility theorem is therefore best understood as specific to a particular market microstructure (competitive, price-taking, no designated market maker), not as a universal statement about all trading mechanisms.

11. Further Literature and Literature Connections

HAYEK (1945), "THE USE OF KNOWLEDGE IN SOCIETY"

Shared idea: the price system as an aggregator of dispersed information. **Relationship:** Hayek is the philosophical ancestor cited directly in the paper's Section IV; Grossman and Stiglitz formalize, in a specific stochastic model, exactly the claim Hayek made informally — that prices economize on the information any individual needs to hold. **Why it matters:** the paper's critique of the efficient markets hypothesis is explicitly framed as a defense of Hayek's insight against a naive reading of "fully reflect," since a price system is valuable in Hayek's sense only when it is aggregating costly, dispersed information — which requires it to be imperfectly revealing.

FAMA (1970), "EFFICIENT CAPITAL MARKETS: A REVIEW OF THEORY AND EMPIRICAL WORK" (ALREADY REVIEWED IN THIS LIBRARY, FINANCIAL MARKETS SERIES)

Shared idea: whether and in what sense prices "fully reflect" available information. **Relationship:** direct target of critique. Grossman and Stiglitz argue Fama's own text (cited at p. 387 of Fama 1970) already recognizes that costless information is *sufficient* for full reflection, but was not aware that positive information cost is what makes full reflection *impossible* to sustain as an equilibrium. **Why it matters:** this is a rare case in the empirical-asset-pricing literature of a purely theoretical paper directly reinterpreting the logical foundations of an already-canonical empirical framework;

reading the two papers together clarifies that Fama's three-way (weak/semi-strong/strong) taxonomy of information sets says nothing about *why* a given information set would or would not be reflected in price, a gap Grossman-Stiglitz fills.

LUCAS (1972), "EXPECTATIONS AND THE NEUTRALITY OF MONEY"

Shared idea: the noisy rational expectations equilibrium concept itself. **Relationship:** Lucas introduced the general technique — agents forming rational expectations conditional on an endogenous, partially revealing signal — that Grossman and Stiglitz apply specifically to a market for costly information. **Why it matters:** the paper is best classified as an application and extension of the Lucas island model's information-theoretic machinery to a genuinely financial-market setting with an explicit market for information itself.

GROSSMAN (1975, 1976, 1977, 1978), VARIOUS

Shared idea: noisy rational expectations equilibria in asset and futures markets, with random supply as the source of noise. **Relationship:** direct precursor and self-citation; this paper is Grossman's own earlier single-author framework extended jointly with Stiglitz to endogenize the fraction of informed traders. **Why it matters:** without the earlier papers' demonstration that random supply (rather than heterogeneous risk preference) is the empirically and analytically preferable source of price noise, the present paper's closed-form tractability would not follow.

GREEN (1977), "THE NON-EXISTENCE OF INFORMATIONAL EQUILIBRIA"

Shared idea: nonexistence of fully revealing rational expectations equilibria. **Relationship:** a direct precursor result cited by the authors (footnote 5) as evidence that informational equilibria "may not, in general, exist." **Why it matters:** Grossman and Stiglitz's contribution relative to Green is to show not merely that a fully revealing equilibrium can fail to exist in some abstract setting, but that the failure occurs precisely at the point where markets would otherwise be judged "efficient," giving the nonexistence result direct relevance to the efficient-markets debate.

HELLWIG (1980), "ON THE AGGREGATION OF INFORMATION IN COMPETITIVE MARKETS," JOURNAL OF ECONOMIC THEORY 22(3), 477–498

Shared idea: noisy rational expectations equilibrium with informed and uninformed traders. **Relationship:** a contemporaneous refinement showing that as the number of traders grows large (a "large market" limit distinct from Grossman-Stiglitz's fixed continuum), price can become approximately, but never exactly, fully revealing, softening the sharp binary impossibility into a continuous large-market approximation result. **Why it matters:** Hellwig shows the impossibility is in some sense a small/fixed-market artifact; in sufficiently large markets, near-full-efficiency and a vanishing individual incentive to acquire costly information can coexist without the discontinuity Grossman-Stiglitz identify, since no single trader's information choice materially affects aggregate informativeness.

DIAMOND AND VERRECCHIA (1981), "INFORMATION AGGREGATION IN A NOISY RATIONAL EXPECTATIONS ECONOMY," JOURNAL OF FINANCIAL ECONOMICS 9(3), 221–235; AND VERRECCHIA (1982), "INFORMATION ACQUISITION IN A NOISY RATIONAL EXPECTATIONS ECONOMY," ECONOMETRICA 50(6), 1415–1430

Shared idea: costly information acquisition in a noisy rational expectations economy with multiple, possibly differentially informed signals. **Relationship:** direct extensions generalizing Grossman-Stiglitz's single-signal, homogeneous-information setup to allow heterogeneous private signals across informed traders. **Why it matters:** these papers test whether the core comparative statics (Theorem 4) and the impossibility logic (Theorem 5) survive richer information structures; broadly, they do, reinforcing that the Grossman-Stiglitz mechanism is not an artifact of assuming all informed traders observe the identical signal θ .

ADMATI (1985), "A NOISY RATIONAL EXPECTATIONS EQUILIBRIUM FOR MULTI-ASSET SECURITIES MARKETS," ECONOMETRICA 53(3), 629–658

Shared idea: CARA-normal noisy rational expectations equilibrium. **Relationship:** extends the single-asset Grossman-Stiglitz apparatus to many correlated risky assets, showing cross-asset information spillovers the single-asset model cannot address. **Why it matters:** demonstrates the limits of Grossman-Stiglitz's single-asset intuition; with correlated assets, a signal about one asset can be informative about others, changing how informativeness and the incentive to acquire information interact across a portfolio, a mechanism entirely absent from the original paper.

KYLE (1985), "CONTINUOUS AUCTIONS AND INSIDER TRADING," ECONOMETRICA 53(6), 1315–1335

Shared idea: a monopolistic informed trader facing noise traders and a competitive market maker. **Relationship:** a competing, strategic alternative to the price-taking Grossman-Stiglitz informed trader; the two papers are the twin foundations of the subsequent asymmetric-information microstructure literature, one competitive and one strategic. **Why it matters:** as discussed in Section 9, Kyle's model does not generate a nonexistence result, which sharpens the interpretation of Grossman-Stiglitz's Theorem 5 as tied specifically to the price-taking assumption.

DE LONG, SHLEIFER, SUMMERS, AND WALDMANN (1990), "NOISE TRADER RISK IN FINANCIAL MARKETS" (ALREADY REVIEWED IN THIS LIBRARY, BEHAVIORAL FINANCE SERIES)

Shared idea: noise, in both papers, is what makes a market for informed activity viable rather than merely a nuisance. **Relationship:** a contrasting later use of "noise": in Grossman-Stiglitz, exogenous supply noise is a technical device that permits equilibrium to exist at all by keeping price imperfectly revealing; in De Long-Shleifer-Summers-Waldmann, noise trader beliefs are themselves a priced source of risk that limits rational arbitrageurs, generating persistent mispricing rather than a well-behaved information equilibrium. **Why it matters:** read together, the two papers show that "noise" plays structurally different, not interchangeable, roles across the rational and behavioral branches of the post-1980 finance literature, despite superficially similar vocabulary.

12. Research Gaps

PAPER-LEVEL GAPS

The welfare properties of the equilibrium are explicitly left open by the authors. The general conjectures of Section I are not proved outside the CARA-normal example. The effect of introducing heterogeneous endowments (footnote 9) on the "vanishing trade" resolution of Theorem 6 is asserted to be a mere complication, not analyzed. The precise threshold $\bar{n}$ governing the sign of $\partial\lambda/\partial n$ (Theorem 4D) is defined only implicitly, with no closed-form characterization or economic interpretation offered in the text.

LITERATURE-LEVEL GAPS

Even after Hellwig (1980), Diamond and Verrecchia (1981), Verrecchia (1982), and Admati (1985), the literature descending from this paper has not, to the reviewer's knowledge, produced a fully general (non-CARA-normal) proof of the seven original conjectures of Section I; tractability has been purchased, across four decades of follow-on papers, largely by retaining the same exponential-normal machinery. Comparatively little of the noisy-rational-expectations literature has been reconciled empirically with the strategic microstructure literature descended from Kyle (1985); the two branches largely coexist rather than nesting one inside the other.

METHODOLOGICAL GAPS

Better data would mean, concretely, direct proxies for the model's primitives — a measurable analogue of the fraction of "informed" market participants (e.g., registered analysts, algorithmic trading desks, or options-implied information shares) and a measurable analogue of price informativeness (e.g., variance ratios, price-efficiency measures from the market microstructure literature) matched against plausible proxies for information cost (e.g., data-vendor subscription prices, regulatory disclosure costs). No such matched dataset is used or proposed by the paper itself, and constructing one remains a live empirical opportunity rather than a solved problem in the literature the reviewer is aware of.

13. Research Opportunities

Idea 1 — Testing the Compensating-Adjustment Prediction with Modern Data Disclosure Shocks

RESEARCH QUESTION: Does a regulatory shock that reduces the cost of acquiring firm-level information (e.g., mandated XBRL structured-data filing, or expanded EDGAR full-text search) leave equilibrium price informativeness roughly unchanged while increasing measured analyst/institutional coverage, as Theorem 4(A)–(B) predicts?

MOTIVATION: Theorem 4's compensating-adjustment result is the paper's sharpest, least intuitive prediction and has, to the reviewer's knowledge, rarely been tested directly against a clean cost-of-information shock.

HYPOTHESIS: around the disclosure-cost reduction, the growth in the number of covering analysts/institutions (a proxy for λ) should be larger, and the change in price informativeness smaller, than a naive model without the endogenous-entry margin would predict.

ECONOMIC MECHANISM: lower c draws in more informed participants until the marginal informed trader is again indifferent, per Theorem 3 and 4; if the market is already well inside the interior-equilibrium region, informativeness gains should be limited relative to the entry response.

LITERATURE CONNECTION: Grossman and Stiglitz (1980), Theorem 4; the disclosure-cost literature (e.g., SEC filing-format mandates).

DATA: SEC EDGAR filing-format transition dates; I/B/E/S analyst coverage counts; institutional 13F holdings counts as a coverage proxy; intraday price-variance-ratio or PIN-style informativeness measures around the same firms.

VARIABLES: analyst/institutional coverage count (proxy for λ), a price-informativeness measure (variance ratio, or probability of informed trading), firm size and industry controls.

METHOD: difference-in-differences or event-study design around the staggered mandate rollout, following firms' coverage and informativeness before and after.

IDENTIFICATION STRATEGY: staggered regulatory rollout across filer size categories as a source of quasi-exogenous variation in information cost, net of confounding firm-level shocks.

EXPECTED CONTRIBUTION: the first direct empirical test, to the reviewer's knowledge, of the specific compensating-adjustment comparative static rather than the more commonly tested general claim that disclosure improves informativeness.

MAIN RISK: coverage counts are an imperfect empirical proxy for the model's binary "informed/uninformed" partition, and disclosure mandates may change signal quality π simultaneously with cost c , confounding the two channels Theorem 4 treats separately.

Idea 2 — Re-Deriving the Model Under Strategic (Kyle-Style) Informed Trading to Isolate the Source of Nonexistence

RESEARCH QUESTION: Does the Theorem 5 nonexistence result survive if informed traders are modeled as a finite number of price-impact-aware strategic agents rather than a continuum of price-takers, as the number of strategic agents grows large?

MOTIVATION: Section 9 above identifies price-taking as the single most consequential assumption behind the impossibility result; this idea tests that diagnosis directly rather than asserting it by comparison across two structurally different papers.

HYPOTHESIS: as the number of strategic informed traders grows without bound, the equilibrium converges to the Grossman-Stiglitz competitive fixed point and inherits its nonexistence at the zero-noise/perfect-information limit; at any finite number of strategic traders, however, equilibrium exists everywhere Grossman-Stiglitz does not.

ECONOMIC MECHANISM: a strategic trader internalizes that his own trade reveals information, moderating trading intensity in a way a price-taker cannot; this endogenous moderation is conjectured to be exactly what restores existence at the limits.

LITERATURE CONNECTION: Grossman and Stiglitz (1980); Kyle (1985); the large-market strategic trading literature.

DATA: none required; this is a theoretical/computational project.

VARIABLES: not applicable (model parameters: number of strategic traders, noise variance, signal precision, cost).

METHOD: Bayesian Nash equilibrium characterization with a finite number of Kyle-style informed traders and endogenous entry, solved numerically for the equilibrium fraction/number entering as a function of the model's noise and precision parameters, taking the large-N limit computationally.

IDENTIFICATION STRATEGY: not applicable; the project is a theoretical convergence result, evaluated by internal proof and numerical limit-taking rather than by identification against data.

EXPECTED CONTRIBUTION: would settle, rather than merely suggest by comparison, whether the impossibility theorem is a property of costly information as such or specifically of competitive (price-taking) behavior.

MAIN RISK: the entry margin (choosing whether to become one of the finite strategic informed traders) is technically difficult to close in a fully strategic setting; the project could stall at a partial characterization without reaching the general convergence result sought.

Idea 3 — Testing Theorem 6's "Vanishing Trade" Resolution with Heterogeneous Endowments Restored

RESEARCH QUESTION: Does trading volume still collapse to zero at the model's degenerate limits once traders are allowed heterogeneous initial endowments of the risky asset, contrary to the pure belief-divergence version the paper solves?

MOTIVATION: footnote 9 of the paper explicitly excludes heterogeneous endowments as an "undue complication," leaving Theorem 6's resolution of the impossibility (markets thin out rather than break down abruptly) untested against a more realistic source of persistent trade.

HYPOTHESIS: with heterogeneous endowments correlated with aggregate supply noise, trade volume remains bounded away from zero even as informativeness approaches its Theorem 5 limits, weakening the "thinness" resolution and sharpening the practical relevance of the underlying nonexistence problem.

ECONOMIC MECHANISM: endowment heterogeneity provides a second, non-informational reason to trade that does not vanish merely because beliefs converge, unlike the pure information-divergence channel modeled in the original paper.

LITERATURE CONNECTION: Grossman and Stiglitz (1980), Section III and footnote 9; Admati (1985) for the multi-asset extension of the same apparatus.

DATA: none required for the core theoretical result; a follow-on empirical test could use portfolio-level 13F endowment dispersion around periods of unusually low measured price informativeness.

VARIABLES: endowment dispersion, per-capita net trade, price informativeness.

METHOD: closed-form re-derivation of Theorems 1–6 with an individual-specific endowment term correlated with, but not identical to, aggregate supply noise, following the CARA-normal solution technique of Appendix B of the original paper.

IDENTIFICATION STRATEGY: theoretical (closed-form re-derivation); an empirical extension would require an identification strategy isolating endowment-driven from information-driven trade, which is a nontrivial further step.

EXPECTED CONTRIBUTION: directly closes the gap the authors themselves flag in footnote 9, and clarifies whether the "vanishing trade" resolution is a general feature of noisy rational expectations models or an artifact of assuming trade is driven by information alone.

MAIN RISK: introducing endowment heterogeneity risks destroying the closed-form tractability that makes the original model solvable, since it reintroduces cross-sectional wealth effects the CARA-normal specification was chosen specifically to avoid.

Idea 4 — Empirically Proxying the Model's Parameters Using Options-Implied Information Shares

RESEARCH QUESTION: Can changes in options-market information share (the fraction of price discovery occurring in the options market relative to the underlying) be mapped onto the model's λ and used to test whether observed comparative statics around known information-cost shocks (e.g., earnings-announcement information environment changes) match Theorem 4's predicted signs?

MOTIVATION: the paper's parameters are unobservable directly, but options markets provide a well-studied, continuously updated proxy for the relative presence of informed trading that could operationalize λ without requiring survey or registration data.

HYPOTHESIS: around a reduction in information cost (e.g., extended trading-halt-free disclosure windows), the options-implied information share rises while the incremental gain in underlying-market price informativeness is smaller than a model without an endogenous entry margin would predict.

ECONOMIC MECHANISM: same compensating-adjustment logic as Idea 1, but using an options-market microstructure proxy for λ rather than a coverage-count proxy.

LITERATURE CONNECTION: Grossman and Stiglitz (1980); the options information-share literature (e.g., Easley, O'Hara and Srinivas-style informed options trading studies).

DATA: OptionMetrics implied volatility and open interest, CRSP/TAQ intraday equity prices, earnings announcement calendars.

VARIABLES: options information share (proxy for λ), equity price variance ratio (informativeness), days-to-earnings (cost/urgency proxy).

METHOD: panel regression of information share and price informativeness around earnings announcements, exploiting cross-sectional variation in analyst coverage and options liquidity as sources of information-cost variation.

IDENTIFICATION STRATEGY: within-firm, around-announcement variation, controlling for firm and event fixed effects; using pre-scheduled earnings dates to avoid reverse causality from price movements to disclosure timing.

EXPECTED CONTRIBUTION: a market-microstructure-based empirical bridge between the Grossman-Stiglitz theoretical apparatus and the well-developed options information-share literature, neither of which is commonly connected to the other in the reviewer's reading of the literature.

MAIN RISK: options information share reflects strategic (Kyle-style) rather than competitive informed trading, so a poor empirical fit to Theorem 4's predictions could reflect model mismatch (Idea 2's concern) rather than a failure of the compensating-adjustment logic itself.

Idea 5 — Extending the Impossibility Result to On-Chain, Fully Transparent Markets

RESEARCH QUESTION: In a blockchain-based market with a fully public, real-time order book and transaction history (near-zero cost of observing all past order flow), does the Grossman-Stiglitz mechanism predict that price cannot be more informative than a positive-information-cost equilibrium permits, and is this testable using on-chain decentralized-exchange data?

MOTIVATION: decentralized exchanges are the closest real-world approximation to a market with near-zero cost of observing aggregate order flow (though not of interpreting it), providing a novel empirical setting to test whether the paper's mechanism operates when the traditionally costly component of "information" shifts from observing trades to interpreting them.

HYPOTHESIS: even with order-flow observation costs near zero, a positive residual cost of interpreting or acting on that flow (e.g., the cost of running predictive models on mempool data) reproduces an analogous equilibrium fraction of "informed" (sophisticated) traders and an analogous cap on price informativeness, consistent with the paper's logic applied to a redefined cost variable.

ECONOMIC MECHANISM: the model's cost parameter c need not represent the cost of observing raw data; it can represent the cost of any activity (e.g., quantitative modeling of mempool data) that converts raw, freely observable information into a private trading edge, preserving the core mechanism in a technologically transparent setting.

LITERATURE CONNECTION: Grossman and Stiglitz (1980); the market microstructure literature on decentralized exchanges and maximal extractable value (MEV).

DATA: on-chain decentralized exchange transaction and mempool data (e.g., Uniswap, via public blockchain explorers and MEV-monitoring datasets).

VARIABLES: a proxy for the fraction of "sophisticated" (modeling-cost-paying) traders, realized price informativeness relative to subsequent price movements, and a proxy for modeling/interpretation cost (e.g., compute cost of competitive MEV strategies).

METHOD: reduced-form analysis of informativeness and sophisticated-trader participation around events that plausibly shift interpretation cost (e.g., release of open-source MEV-detection tooling).

IDENTIFICATION STRATEGY: event-study design around tooling releases that lower interpretation cost for a broad set of participants simultaneously, analogous to Idea 1's regulatory-shock design.

EXPECTED CONTRIBUTION: would test whether the Grossman-Stiglitz mechanism is genuinely about the cost of any information-processing activity, as its internal logic suggests, or specifically tied to a pre-digital notion of information scarcity that a fully transparent, blockchain-based market might render moot.

MAIN RISK: decentralized-exchange markets differ from the paper's assumptions in several first-order ways (block-time discreteness, atomic transaction ordering, public mempools enabling front-running) that could dominate any Grossman-Stiglitz-style effect and confound identification.

Research Idea Ranking

Ranked by a combination of novelty, feasibility, data availability, methodological quality, and expected economic importance: (1) **Idea 1** — the most feasible with existing, high-quality public data (EDGAR, I/B/E/S, 13F) and a clean regulatory shock for identification, directly testing the paper's sharpest and least-tested prediction; (2) **Idea 2** — the highest-value theoretical

contribution, since it would resolve an open interpretive question (Section 9's central critique) that has stood since Kyle (1985) without, to the reviewer's knowledge, being directly closed; (3) **Idea 4** — strong data availability (OptionMetrics, TAQ) and a well-established complementary literature, though weaker identification than Idea 1 because options information share is itself a somewhat strategic, not purely competitive, construct.

Ideas 3 and 5 rank lower on feasibility: Idea 3 risks losing closed-form tractability (a real and non-trivial technical risk, per the paper's own footnote 9), and Idea 5 faces a market microstructure environment different enough from the model's assumptions that a clean test is harder to design, even though its novelty and long-run importance for understanding transparent, blockchain-based markets are high.

14. Learning Framework

PREREQUISITE CONCEPTS

Constant absolute risk aversion (CARA) utility and its mean-variance-equivalent demand functions; properties of the multivariate normal distribution, especially conditional expectation and conditional variance under joint normality (Appendix A of the paper); the basic rational expectations equilibrium concept (agents condition on an endogenous signal — here, price — and their beliefs about its informational content are correct in equilibrium); and the concept of a fixed point (here, in the fraction of informed traders) as the solution concept for a market that clears in two dimensions simultaneously (the asset market and the market for information).

CORE CONCEPTS LEARNED

That price informativeness and the private return to information are jointly, not independently, determined; that a competitive equilibrium requires a source of noise to remain non-degenerate whenever information is costly; that "efficiency," properly defined, is a matter of degree pinned down by a small number of structural parameters (cost, signal quality, risk aversion) rather than a binary property a market either has or lacks; and that the limiting, fully efficient case is not merely rare but strictly impossible to sustain as an equilibrium under the model's assumptions.

IMPORTANT EQUATIONS

Equation (8), the informed demand function, as the building block of every subsequent result; equation (19), the scalar equilibrium condition $e^{2ac} = 1 + nm/(1+m)$, as the single equation from which all comparative statics follow; and the two conditions of Theorem 5 as the precise mathematical statement of the paper's title result.

METHODS LEARNED

Solving for a linear rational expectations equilibrium price function by conjecture-and-verify (Theorem 1's proof technique); computing conditional moments under joint normality using the projection formulas of Appendix A; reducing a multi-parameter comparative-statics problem to a

single composite ratio (m, n) to make monotonicity arguments tractable; and using a discontinuity-at-a-boundary argument to prove nonexistence, rather than attempting (impossible, in this case) to exhibit a counterexample equilibrium directly.

QUESTIONS TO TEST UNDERSTANDING

Comprehension: In your own words, why does full price revelation eliminate the incentive to pay for information, and why does that in turn eliminate full price revelation? **Analysis:** Why does the equilibrium condition (19) depend on noise and total variance only through λ and not directly — what economic force is doing the offsetting? **Research:** If informed traders were modeled as strategic price-setters rather than price-takers, would you expect Theorem 5's nonexistence result to survive, weaken, or disappear, and why?

PAPERS TO READ NEXT

Kyle (1985), "Continuous Auctions and Insider Trading," for the strategic counterpart to this paper's competitive framework; Hellwig (1980), "On the Aggregation of Information in Competitive Markets," for the large-market refinement of the same impossibility; and Milgrom and Stokey (1982), "Information, Trade and Common Knowledge," for the closely related no-trade theorem that sharpens the question of why any trade based purely on private information should occur at all.

15. Conclusion and Final Synthesis

ONE-SENTENCE THESIS

In a competitive market where acquiring information is costly, prices can be informative but can never be fully, perfectly informative, because full informativeness would eliminate the very return to information that makes prices informative in the first place.

THREE MOST IMPORTANT INSIGHTS

First, informational efficiency and the private incentive to produce information are two sides of a single equilibrium condition, not independent market properties. Second, the equilibrium level of price informativeness depends only on the cost of information, the quality of the informed signal, and risk aversion — not on the level of noise, which is instead absorbed by an offsetting adjustment in the fraction of informed traders. Third, the paper's impossibility is a genuine logical limit, not a statement about how close to efficient real markets typically are; it identifies a boundary that a costly-information competitive market can approach but never reach.

STRONGEST EVIDENCE

The closed-form derivation of the scalar equilibrium condition (19a), from which Theorems 4, 5, and 6 all follow by direct algebraic manipulation, is the paper's strongest and most rigorously supported result — a fully specified, provably unique equilibrium characterization rather than an informal argument.

BIGGEST WEAKNESS

The reliance on price-taking, non-strategic informed traders, whose absence of price impact is precisely what forces the sharp discontinuity behind Theorem 5; Kyle's (1985) strategic alternative shows the nonexistence result does not survive once informed traders are allowed to perceive their own effect on price.

MOST IMPORTANT RESEARCH GAP

No empirical study, to the reviewer's knowledge, has directly tested the compensating-adjustment prediction of Theorem 4 — that equilibrium price informativeness is invariant to the level of noise trading holding cost, signal quality, and risk aversion fixed — against a real cost-of-information shock.

BEST RESEARCH OPPORTUNITY

Idea 1: using a staggered regulatory disclosure-cost shock (such as structured-data filing mandates) as a natural experiment to test Theorem 4's compensating-adjustment prediction directly, using existing, high-quality public coverage and informativeness data.

THREE PAPERS TO READ NEXT

Kyle (1985), "Continuous Auctions and Insider Trading"; Hellwig (1980), "On the Aggregation of Information in Competitive Markets"; and Admati (1985), "A Noisy Rational Expectations Equilibrium for Multi-Asset Securities Markets."

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